

INT ZONE STUDIO

Complete User Guide & Reference Manual



An engineering manual detailing automatic slab polygonization, concave boundary clipping, validation-driven gap seal recovery, and concrete takeoff deliverables for structural and civil design.

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Release Version:	0.1.0-pilot.1 (Standalone Windows Local)
Publisher:	INT Zone Studio Team

Pilot Evaluation Build v1 · Version **0.1.0-pilot.1**

This complete reference guide walks you through the desktop CAD workspace, explaining every menu ribbon button, sidebar panel, canvas mouse control, and step-by-step quantity takeoff workflow.

Chapter 1: Welcome & Overview

INT Zone Studio is a desktop CAD engineering application designed to automate concrete slab quantity takeoffs. Instead of spending hours manually tracing perimeters in AutoCAD, the geometric engine parses your foundation drawing layers to automatically detect closed areas, highlight slab boundary gaps, and generate concrete area/volume reports.

1.1 System Requirements

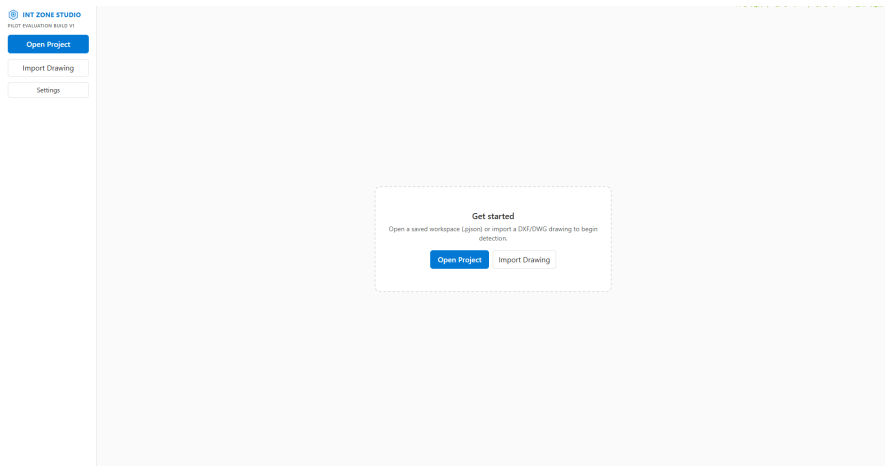
Component	Minimum Specification	Recommended Specification
OS	Windows 10 (64-bit)	Windows 11 (64-bit)
RAM	8 GB	16 GB or higher
Processor	Intel Core i5 / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
Graphics	Integrated Intel/AMD	Dedicated GPU (Nvidia/AMD)
Python	Not Required	Not Required (built-in sidcar engine)

1.2 Installation & Startup

1. Download **INT Zone Studio Standalone Setup 0.1.0-pilot.1.exe** from GitHub Releases.
2. Double-click to run the setup wizard and complete the installation.
3. Open the app from your **Start Menu** or Desktop shortcut.

1.3 First Launch Checklist

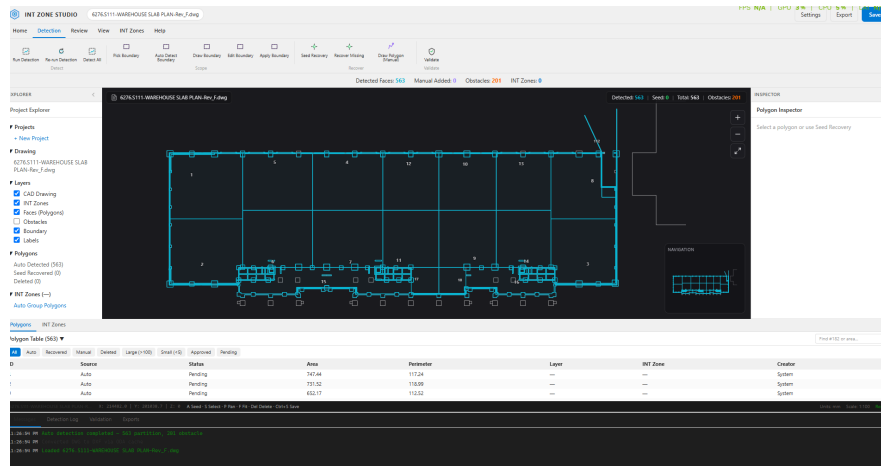
1. **Wait 5 Seconds:** After launching the app, wait 5 seconds for the background geometry engine to establish a connection.
2. **Select Start Option:** You will be greeted with the Welcome Screen:
 - **Import Drawing:** Click to select a new AutoCAD **.dxf** or **.dwg** file.
 - **Open Project:** Click to select a previously saved **.pjson** workspace file.



Welcome Screen on Startup

Chapter 2: Interactive Viewport Navigation

The CAD Canvas is your main viewport. Navigating and interacting with the drawing is fast and responsive thanks to mouse scroll and key bindings.



Workspace Canvas Overview

2.1 Viewport Panning (Moving Around)

- **Spacebar Panning (Recommended):** Press and hold the **Spacebar** key on your keyboard. The cursor turns into a hand icon. Left-click and drag anywhere on the canvas to move the drawing. Releasing the Spacebar exits pan mode and returns to your active tool.
- **Pan Tool:** Alternatively, click the **Pan (P)** button on the View tab to lock pan mode.

2.2 Viewport Zooming (Resizing)

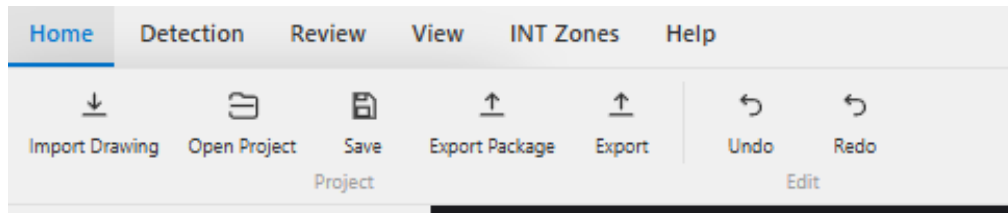
- **Mouse Scroll Wheel (Recommended):** Scroll the wheel **Up** to zoom in, and scroll **Down** to zoom out. Zooming centers dynamically on the exact position of your mouse pointer.
- **Zoom Window (Z):** Click the **Zoom Window** tool on the View tab, then click and drag a rectangular window on the canvas to zoom into that region.
- **Floating Controls:** Click the floating **Plus (+)**, **Minus (-)**, or **Fit View (Maximize)** buttons in the bottom right corner of the canvas.

2.3 Mouse Click Interactivity

- **Left-Click Selection:** Click any slab polygon on the canvas (when the **Select (S)** cursor is active) to view its area details in the Properties sidebar.
- **Shift + Click:** Select multiple polygons at once.
- **Drawing Vertices:** Left-click to place nodes while drawing boundaries or manual polygons.
- **Right-Click (Undo Vertex):** While drawing a line or boundary manually, right-click to delete the last vertex.
- **Double-Click / Enter:** Double-click or press Enter to close the loop (joins last node to first node) and finish drawing.
- **Escape (Esc):** Cancels active previews, active drawings, or selections.

Chapter 3: Detection Tab & Boundary Operations

The **Detection** tab is the core control center of the geometric analysis engine. It contains tools to run polygon extraction, outline perimeters, and restore missing slabs.



Detection Tab Ribbon

3.1 The "Detect" Group (Face Extraction)

- **Run Detection (D):** Instructs the engine to read the active drawing layers and calculate closed loops formed by lines, polylines, and arcs. It automatically builds "faces" (cyan polygons) wherever a space is fully enclosed.
- **Re-run Detection:** Forces the canvas to redraw polygons. This does not reload the DXF file from disk, meaning manual reviews, approvals, and recoveries are preserved.
- **Detect All:** A batch command that overrides active layer filters to trace closed loops on every single layer in the drawing.

3.2 The "Scope" Group (Slab Boundaries)

Defining a boundary restricts the area of calculation to the actual concrete slab layout. This is crucial for filtering out outdoor paving, yard spaces, or surrounding structural noise.

How to Define a Boundary (3 Methods) Choose one of these tools under the **Scope** group to create a boundary candidate:

1. **Pick Boundary (B):** Select this tool, then click on an existing closed AutoCAD polyline on the canvas. Use this if the CAD operator has already drawn the slab perimeter.
2. **Auto Detect Boundary:** The engine calculates a concave hull wrapping around the outer wall elements.
3. **Draw Boundary:** Left-click on the canvas to place points manually around the perimeter. Hovering near corners activates the snap helper. Double-click or press Enter to close the boundary path.

Confirming and Editing the Boundary

- Once defined, a yellow dashed perimeter preview will appear. Click **Accept** in the top black banner to save it.
- To make adjustments, click **Edit Boundary**. Click and drag any yellow node to move it. Click along an edge line to add a new vertex node. Right-click or press Delete on a selected node to remove it.

The Crucial Step: Apply Boundary

- > [!IMPORTANT]
 - **Simply drawing or picking a boundary does not crop your drawing.**
 - To apply the filter, you **must** click the **Apply Boundary** button.
 - Once clicked, the engine filters all polygons. Any polygon whose centroid lies outside the blue boundary is marked as "excluded" and disappears from the viewport and schedule.

3.3 The "Recover" Group (Resolving Gaps)

When CAD draftsmanship is imperfect, walls do not touch, leaving gaps. The engine cannot detect a closed loop. Use these recovery tools to fix them:

- **Seed Recovery (R):**
 - Select the tool, then zoom into the open room area.
 - Click in the center of the room.

- The engine shoots virtual ray sensors outward. If it finds a gap smaller than the threshold tolerance, it automatically bridges the gap and highlights a green preview polygon.
- Click **Accept** in the top banner to save the recovered polygon.
- **Recover Missing:** Click this while a suspected gap coordinate is highlighted in the Validation list to instantly construct a slab polygon at that gap's center.
- **Draw Polygon (Manual) (M):** If a wall is entirely missing from the drawing, click around the corners to manually draw a custom polygon boundary. Double-click to close and click **Accept** in the banner.

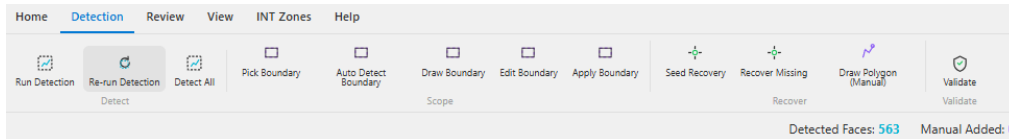
3.4 Validation and Statistics Groups

- **Validate (V):** Traces drawing junctions to locate line endpoints that do not intersect. The coordinates are exported to the Suspected Gaps panel.
- **Statistics:** Displays metadata details: **Detected Count** (number of slab cells), **Missing Count** (gaps remaining), and **Coverage %** (percentage of slab area correctly covered by approved polygons).

Chapter 4: Tab Ribbon & Panels Reference

This chapter details the remaining menu tabs, buttons, sidebars, and panels that make up the workspace.

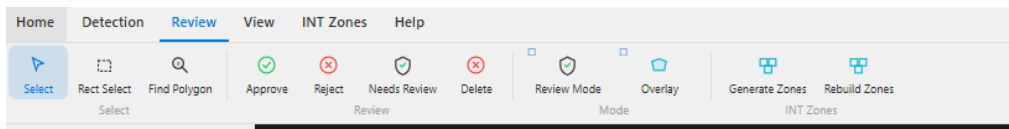
4.1 Home Tab (Project & Export)



Home Tab Ribbon

- **Import Drawing:** Opens a DXF/DWG file selector. Large layouts take **30-60 seconds** to calculate.
- **Open Project (Ctrl+O):** Restores a saved **.pjson** project.
- **Save (Ctrl+S) & Save Workspace As:** Saves the active session. Type or paste the full file path (e.g. **C:\Projects\SlabSectionA.pjson**) into the text box.
- **Export Package:** A single click generates a dedicated folder containing the Excel takeoff schedule, annotated DXF drawing, and PDF summary.
- **Export:** Individual settings to export DXF only or Excel sheet only.
- **Undo (Ctrl+Z) / Redo (Ctrl+Y):** Viewport transaction controls.

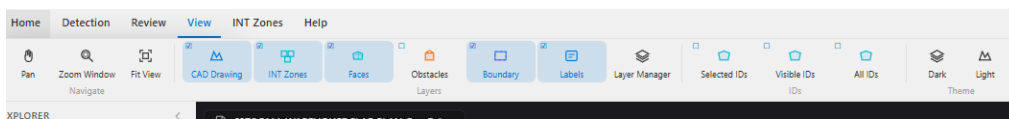
4.2 Review Tab (Auditing & Polygons)



Review Tab Ribbon

- **Select (S):** Activates the selection pointer.
- **Rect Select:** Click-and-drag a box to select multiple polygons.
- **Find Polygon (Ctrl+G):** Locate a polygon by ID.
- **Approve / Reject / Needs Review / Delete (Del):** Polygon audit status buttons.
- **Toggle Review Mode:** Displays properties table and validation dock.
- **Comparison Overlay:** Highlights auto-detection differences.
- **Generate Zones:** Aggregates detected rooms into INT zones.

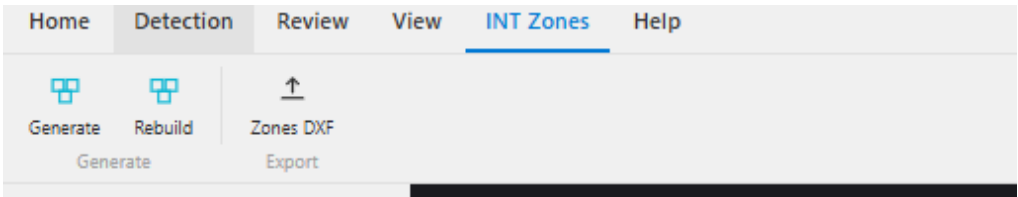
4.3 View Tab (Layers & Workspace Panels)



View Tab Ribbon

- **Pan (P) / Zoom Window (Z) / Fit View (F):** Viewport shortcuts.
- **Layers Group:** Toggle checkboxes to show or hide: **CAD Drawing**, **INT Zones**, **Faces**, **Obstacles**, **Boundary**, and **Labels**.
- **IDs Group:** Show IDs for **Selected**, **Visible**, or **All** polygons.
- **Theme Group:** Toggle between **Dark** and **Light** modes.

4.4 INT Zones Tab (Aggregations)

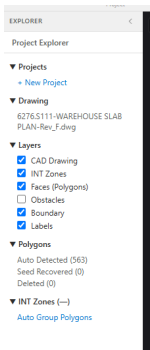


INT Zones Tab Ribbon

- **Generate / Rebuild:** Groups smaller polygons into INT layout zones.
- **Analysis:** Shows zone concrete statistics and areas.
- **Export:** Exports the INT-specific schedule DXF.

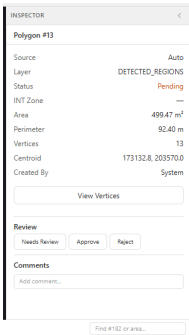
4.5 Sidebar & Dock Panels

- **Polygon Table (Left Sidebar):** Displays the full list of detected slab areas. Selecting a row pans the canvas directly to that polygon.



Polygon Table Sidebar

- **Properties Panel (Right Sidebar):** Displays detailed dimensions (Area m², Concrete Thickness m, and Volume m³) and status buttons for the selected polygon.



Properties Panel Sidebar

- **Validation & Gaps Panel (Right Bottom):** Lists all open corner disconnects and line gaps. Double-click a gap entry to center the canvas on it, then use Seed Recovery to close the gap.

ID	Auto	Recommended	Manual	Deleted	Large (>100)	Small (<5)	Approved	Pending	Area	Perimeter	Layer	INT Zone	Creator
9								Pending	122.37	96.27	—	—	System
10								Pending	119.27	94.55	—	—	System
11								Pending	114.15	95.12	—	—	System
12								Pending	109.68	93.40	—	—	System
13								Pending	499.47	92.40	—	—	System
14								Pending	496.66	101.63	—	—	System
15								Pending	476.56	236.10	—	—	System
16								Pending	247.21	127.89	—	—	System
17								Pending	112.41	44.45	—	—	System
18								Pending	110.21	42.95	—	—	System
19								Pending	64.39	239.82	—	—	System
20								Pending	30.51	109.91	—	—	System
21								Pending	21.48	19.79	—	—	System
22								Pending	13.43	95.22	—	—	System
23								Pending	12.89	81.51	—	—	System
24								Pending	12.31	81.15	—	—	System

Validation and Statistics Bottom Panel

Chapter 5: Step-by-Step Quantity Takeoff Workflow

Follow this standard engineering session sequence:

1. **Import Drawing:** Click **Home** → **Import Drawing** and load your DXF/DWG file.
2. **Detect Polygons:** Click **Detection** → **Run Detection** (or press **D**). Cyan polygons appear.
3. **Set Perimeter:** Go to **Detection Tab**. Select **Pick Boundary** and click the slab outer line. Click **Accept** in the top banner, then click **Apply Boundary**.
4. **Find Wall Gaps:** Click **Review** → **Run Validation** (or press **V**). Open the **Suspected Gaps** list.
5. **Seal Gaps:** Double-click a gap entry in the list to center the view. Click **Detection** → **Seed Recovery** (or press **R**), click inside the room, and click **Accept** in the top banner.
6. **Review & Approve:** Click **Select (S)**, select polygons, check areas, and mark them **Approved** (or press **Del** to remove invalid shapes).
7. **Save Project:** Press **Ctrl+S**, write a filename path (e.g. **C:\Slabs\Session1.pjson**), and click Save.
8. **Export Deliverables:** Click **Home** → **Export Package**. Check the output folder for your finalized Excel quantity takeoff schedule, annotated DXF drawing, and PDF summary.

Chapter 6: Keyboard Shortcuts Quick Reference

Key / Shortcut	Mode	Description
Spacebar (Hold)	Canvas navigation	Hold + Left-Click & Drag to pan the drawing
Mouse Scroll	Canvas navigation	Zoom in (Scroll Up) / Zoom out (Scroll Down) centered on mouse pointer
Ctrl + O	File Operation	Opens a saved .pjson project
Ctrl + S	File Operation	Saves the active workspace session
Ctrl + Z	Edit	Undo the last action
Ctrl + Y	Edit	Redo the last action
Ctrl + G	Selection	Find a polygon by typing its ID number
D	Detection	Triggers automatic slab detection
B	Scope	Pick an AutoCAD polyline as boundary
R	Recovery	Activates the Seed Recovery (click-to-close gap) tool
M	Recovery	Activates Draw Polygon (Manual) tool
V	Validation	Scans drawing and displays suspected gaps
S	Selection	Switches to Select mode
P	Canvas Navigation	Toggles pan tool lock
Z	Canvas Navigation	Toggles Zoom Window rectangle tool
F	Canvas Navigation	Fits entire drawing to screen
Del	Edit	Deletes the selected polygon
Esc	Global	Cancels active drawing, seed preview, or selection

Chapter 7: FAQs & Troubleshooting

Q1: The import progress bar is stuck. What should I do?

- **Reason:** The CAD drawing is extremely complex or contains corrupt block references.
- **Fix:** Wait up to 60 seconds. If it remains stuck, close the app, open your AutoCAD file, save it as a DXF format, and import the DXF.

Q2: Seed Recovery is not creating a polygon.

- **Reason:** The gap in the walls is too large to auto-close, or your click was too close to a wall line.
- **Fix:** Zoom in close. Make sure your click is in the absolute center of the room. If it still fails, use the **Draw Polygon (Manual) (M)** tool to trace the shape manually.

Q3: How are columns and pillars handled?

- **Fix:** Turn ON the **Obstacles** layer in the View tab to verify column block insert footprints (orange). Obstacles are automatically identified and marked for review.

Q4: Save Workspace fails with "directory error".

- **Fix:** You must type a folder path that already exists in Windows Explorer (e.g., **C:\Users\Administrator\Desktop\Slab.json**). The pilot build does not create new folders.

INT Zone Studio · Pilot Build Edition